

EE1080/AI1110/EE2102 Probability and Random Processes

Batch 1

Updated: 13 January, 2026

1 Course Platform

We will use google classroom to post homework problems, lecture notes, announcements and discussions. You can enroll using the join code: **3tdv4gst**

2 Time and Venue

The classes will be held every Mon: 12-1pm, Tues: 9-10am and Fri: 11am-12pm at ~~LHC-42~~ A-LH-1
Office hours: Tuesday: 1-2pm

3 Course Overview

1. Basics of Probability
 - probability model, axioms of probability, conditional probability, basics of counting
2. Random Variables
 - discrete and continuous random variables, cdf, expectation, independence, conditional expectation, functions of random variables etc, probabilistic method
3. Concentration Inequalities and Limit Theorems
 - markov type inequalities, convergence of random variables and limit theorems.
4. Inference: estimation with least error with and without prior
5. Gaussian Random Vectors

4 Grading

1. Bi-Weekly Quizzes: 45%
 - worst two will be ignored, rest will be given equal weightage. Tentative schedule for quizzes:
 - (a) Quiz 1: 19th Jan (Monday) 6-7 pm
 - (b) Quiz 2: 2nd Feb (Monday) 6-7 pm
 - (c) Quiz 3: 16th Feb (Monday) 6-7 pm
 - (d) Quiz 4: 23rd March (Monday) 6-7 pm
 - (e) Quiz 5: 6th April (Monday) 6-7 pm
 - (f) Quiz 6: 20th April (Monday) 6-7 pm
2. Exams: 45%
 - (a) Midterm: 25th Feb(Wednesday) ~~2-5~~ 4-7pm
 - (b) Endterm: 30th April (Thursday) 10am-1pm
 - There will be two exams, one mid semester and one towards the end.
3. Programming Assignments: 10%
 - There will be two programming assignments with tentative deadlines on 15th March, 24th April. You will get 10-14 days after the assignment is posted for the submission.
4. Homework: You will be provided practice sheets bi-weekly. They need not be submitted.

5 Expectation from the Student

- Students are expected to attend and participate in all the lectures, quizzes and exams
- Solve all the homework problems on your own
- Do not engage in any unfair academic practices.

6 References

- Introduction to Probability by Bertsekas, Tsitsiklis
- Probability, Random Variables and Stochastic Processes by Papoulis and Pillai
- Introduction to probability for data science by Stanley H. Chan
- First course on probability by Sheldon M. Ross
- List will be updated as the course progresses

7 Teaching Team

- Instructor: Myna Vajha
- TAs:
 - Undergrad TAs: Abhinav Koushik, Agamjot, Akshara Sarma, Faraz, Shubhang, Yashwanth
 - Postgrad TAs: